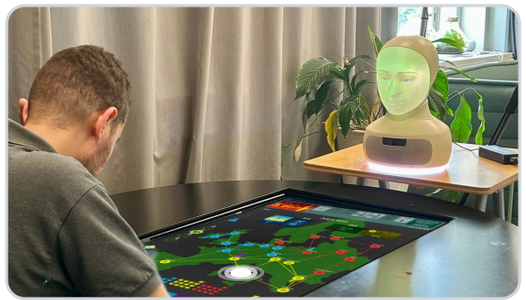
1 Testbed



state ↓ ↑ moves

rule-based solver

HRI corpus
70 sessions · 36 participants
speech · video · gaze · game logs

2 Partner signals

training time only

a Observed
speech
video
game state

b Self-reported
engagement ●●●●○
enjoyment ●●●●●
cognitive effort ●●○○○

c Measured
brain activation
eye tracking
task performance **win**

3 Model adaptation *current/future work*

trained

full-duplex h_t

partner audio
partner gaze
task state
LLM
text head
gaze head
action head
bias
robot speech
robot gaze
robot action

stage 1 · fine-tune
$\pi \cdot v$ self-play
gaze data
***where** · abundant test-time inputs*

80 ms

stage 2 · steer
recurrent intervention head
when · whether

predictor